

SHETH MVLU COLLEGE

SUBJECT:- WSN

Aim: Implementation of nesC Programming Model

Create a basic nesC application using modules and configurations to understand component wiring and interface binding.

Step 1: ([BlinkC.nc](#))

CODE:

```
#include "Timer.h"

module BlinkC {
  // Interfaces used by this module
  uses interface Boot;
  uses interface Leds;
  uses interface Timer<TMilli> as Timer0;
}

implementation {
  event void Boot.booted() {
    // Command: start timer
    call Timer0.startPeriodic(500);
  }

  event void Timer0.fired() {
    // Command: toggle LED
    call Leds.led0Toggle();
  }
}
```

Step 2:([BlinkAppC.nc](#))

Code:

```
configuration BlinkAppC {
}

implementation {
  // 1. Declare the components
  components MainC;
  components LedsC;
  components new TimerMilliC() as Timer0;
  components BlinkC as App;

  // 2. Wire the interfaces (Interface Binding)
  App.Boot -> MainC.Boot;
  App.Leds -> LedsC;
  App.Timer0 -> Timer0;
}
```

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Step 3: (sim.py)

CODE:

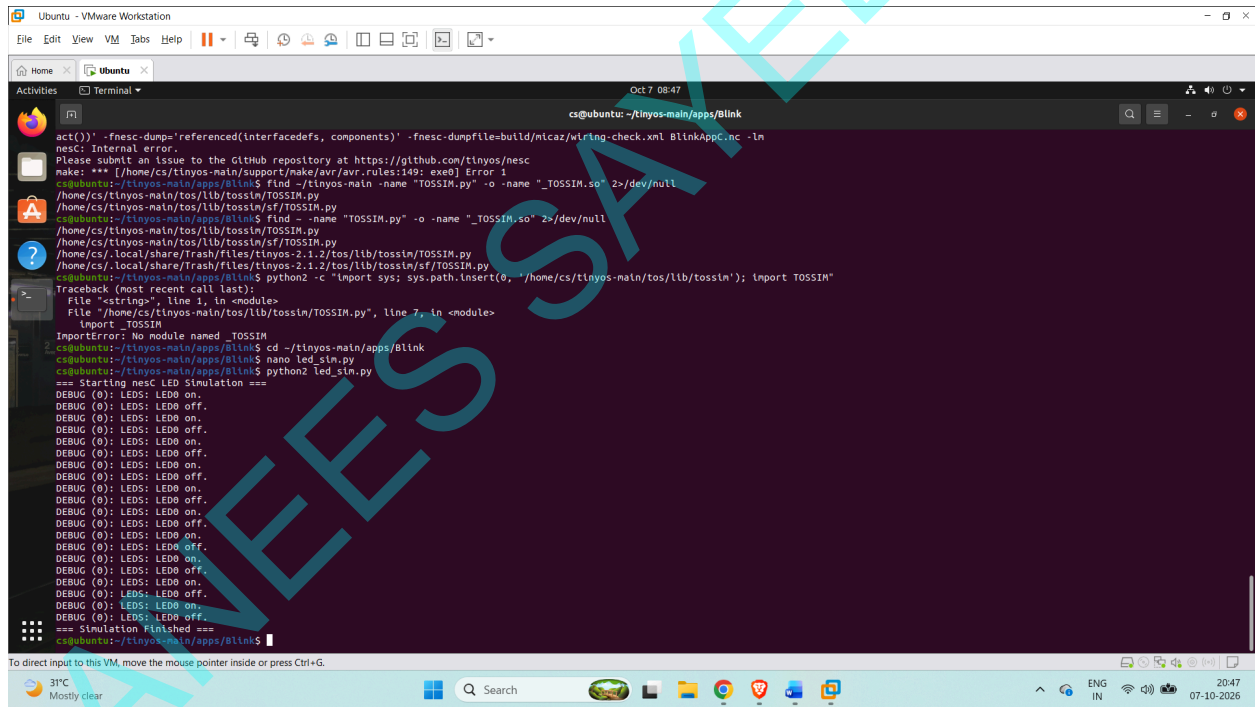
```
import sys
from TOSSIM import *

t = Tossim([])
t.addChannel("LedsC", sys.stdout)

# Boot mote 1
m = t.getNode(1)
m.bootAtTime(0)

# Run 20 events
for i in range(20):
    t.runNextEvent()
```

OUTPUT



```
cs@ubuntu:~/tinyos-main/apps/Blink
act())' -fnesc-dumpfile=build/micaz/wiring-check.xml BlinkApp.nc -ln
nescC: Internal error.
Please submit an issue to the GitHub repository at https://github.com/tinyos/nesc
make: *** [/home/cs/tinyos-main/support/make/avr/avr.rules:149: exe0] Error 1
cs@ubuntu:~/tinyos-main/apps/Blink$ find ~//tinyos-main -name "TOSSIM.py" -o -name "_TOSSIM.so" 2>/dev/null
/home/cs/tinyos-main/tos/lib/tossin/TOSSIM.py
/home/cs/tinyos-main/tos/lib/tossin/sf/TOSSIM.py
cs@ubuntu:~/tinyos-main/apps/Blink$ find -name "TOSSIM.py" -o -name "_TOSSIM.so" 2>/dev/null
/home/cs/tinyos-main/tos/lib/tossin/TOSSIM.py
/home/cs/tinyos-main/tos/lib/tossin/sf/TOSSIM.py
/home/cs/.local/share/Trash/files/tinyos-2.1.2/tos/lib/tossin/sf/TOSSIM.py
cs@ubuntu:~/tinyos-main/apps/Blink$ python2 -c "import sys; sys.path.insert(0, '/home/cs/tinyos-main/tos/lib/tossin'); import TOSSIM"
Traceback (most recent call last):
  File "<string>", line 1, in <module>
  File "/home/cs/tinyos-main/tos/lib/tossin/TOSSIM.py", line 7, in <module>
    import _TOSSIM
ImportError: No module named _TOSSIM
cs@ubuntu:~/tinyos-main/apps/Blink$ cd ~//tinyos-main/apps/Blink
cs@ubuntu:~/tinyos-main/apps/Blink$ nano led_sim.py
cs@ubuntu:~/tinyos-main/apps/Blink$ python2 led_sim.py
=== Starting nesc LED Simulation ===
DEBUG (0): LEDS: LED0 on.
DEBUG (0): LEDS: LED0 off.
DEBUG (0): LEDS: LED0 on.
DEBUG (0): LEDS: LED0 off.
DEBUG (0): LEDS: LED0 on.
DEBUG (0): LEDS: LED0 off.
DEBUG (0): LEDS: LED0 on.
DEBUG (0): LEDS: LED0 off.
DEBUG (0): LEDS: LED0 on.
DEBUG (0): LEDS: LED0 off.
DEBUG (0): LEDS: LED0 on.
DEBUG (0): LEDS: LED0 off.
DEBUG (0): LEDS: LED0 on.
DEBUG (0): LEDS: LED0 off.
DEBUG (0): LEDS: LED0 on.
DEBUG (0): LEDS: LED0 off.
DEBUG (0): LEDS: LED0 on.
DEBUG (0): LEDS: LED0 off.
DEBUG (0): LEDS: LED0 on.
DEBUG (0): LEDS: LED0 off.
=== Simulation finished ===
cs@ubuntu:~//tinyos-main/apps/Blink$
```

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